Juan Marulanda

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Education

Carleton University — Ottawa, ON
Bachelor of Computer Science, AI and Machine Learning Concentration — *2023 – 2027*

- **Relevant Coursework:** Data Analytics, Algorithms and Data Structures, Operating Systems, Compiler Construction, ML

Experience

Tatlock Systems — Ottawa, ON
Software Engineer Intern — *May 2026 – Aug 2026*

- Designing and implementing a safe dialect of C programming language targeting micro-controllers, with safety guarantees at compile time.
- Built an LSP server and VS Code extension for the language, parsing source files to provide syntax highlighting and statement simplification for beginners.
- Collaborating with researchers in the low-latency and high-performance computing industry to validate design decisions and iterate on language semantics

Bedarra Corporation — Ottawa, ON
Software Engineer Intern — *May 2024 – Aug 2025*

- Engineered a local-first cross-platform app (iOS, Android, Web, macOS, Windows) with a CRDT-based sync engine using Iroh, implementing store-and-forward propagation so devices sync opportunistically on shared networks without central coordination.
- Designed a key-namespaced document schema with last-write-wins resolution, syncing metadata immediately and fetching content asynchronously via Iroh's content-addressed blob store across disconnected peers.
- Engineered an on device email ingestion pipeline and local first authentication using UCAN tokens, eliminating server dependencies and improving latency and availability.

Projects

RISC-V Kernel | *C, RISC-V Assembly, QEMU*

- Built a RISC-V kernel from scratch in C, implementing context switching, virtual memory, user mode, a shell, a disk driver, and a file system to produce a fully functional bootable OS.
- Identified and resolved kernel faults by leveraging QEMU's GDB stub and raw memory inspection, cutting crash frequency and delivering a stable system.

Fast Web Browser | *Rust*

- Implemented a high-performance HTML pre-scanner using x86_64 SIMD intrinsics (AVX2) to minimize parsing latency and maximize instruction-level parallelism.
- Engineered a parser that transforms token patterns into finite state machines, streamlining lexical analysis and cutting tokenization time.
- Designed a Structure-of-Arrays (SoA) DOM layout for cache locality, and implemented HTTP/1.1 networking, the TLS handshake, and a software rasterizer from scratch.

Technical Skills

Languages: Rust, C/C++, TypeScript, Haskell, Python, Java, Dart, SQL (PostgreSQL, OracleSQL, SQLite), JavaScript
Frameworks: Flutter, React, Node.js, Tokio, Flask, FastAPI, Deno.js
Developer Tools: Git, Docker, perf, eBPF, Cursor, Claude Code, PostgreSQL, Linux (WSL), Xcode, Android Studio
Technologies: FFI, WASM, P2P Protocols, REST APIs, IMAP, SMTP

Extracurricular Activities

uOttaHack (Student-Led Hackathon)
Community Coordinator

- Helped organize Ottawa's largest annual hackathon (uOttaHack), a Major League Hacking event run in partnership with the University of Ottawa, facilitating workshops, networking, and community-building for 900+ student participants.

Synchronize (Ottawa's Socratica Node)
Host and Organizer

- Host and facilitate weekly co-working sessions for Ottawa's Socratica node, one of 40+ global nodes, bringing together makers, engineers, artists, and designers to work on passion projects and share progress through open demo.